

You are a tutor helping me prepare for UC Berkeley CS 182: Deep Neural Networks, specifically its prerequisite knowledge with HW0 as your reference (attached).

With each prompt, you are going to give 10 high quality and unique Anki-style flashcards that cover the following topics geared towards preparation for CS 182:

Prerequisite topics the course expects, for example:

Linear algebra: vectors, matrices, norms, dot products, eigenvalues/eigenvectors

Probability: random variables, expectation, variance, conditional probability

Calculus: derivatives, gradients, chain rule, basic optimization idea

Python/NumPy: broadcasting, basic tensor ops, shape reasoning

These are flashcard quality rules for you to adhere to:

Active recall, not recognition

Minimum-information principle: one fact per card

No long paragraphs on front or back

Use cloze deletions for formulas or multi-step definitions

where helpful

Avoid vague words (“explain everything about...”)

For this first batch, focus only on linear algebra prerequisites for CS 182 (vectors, matrices, norms, dot products, eigenvalues/eigenvectors).

Annotations look like this!

Produce exactly 10 high-quality flashcards that follow all the rules above.

Here are **10 Anki-style flashcards** focusing only on **linear algebra prerequisites** for CS 182 (vectors, matrices, norms, dot products, eigenvalues/eigenvectors). All cards follow active-recall, minimum-information, and cloze-deletion principles. No content is copied from HW0; this batch only uses HW0 to guide prerequisite selection.  hw0

Flashcards (10 total)

1. Vector norm definition

Front: What is the definition of the ℓ_2 norm of a vector $x \in \mathbb{R}^n$?

Back: $\|x\|_2 = \sqrt{\sum_i x_i^2}$

These are good and match the format and style that I am going for, but the difficulty of the questions is debatable, with some seeming too easy for students who succeeded in all prerequisite classes.

2. Dot product–norm relation

Front: The dot product $x^\top x$ equals what quantity?

Back: $\|x\|_2^2$

3. Matrix–vector product shape

Front: If $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$, what is the shape of Ax ?

Back: $m \times 1$

4. Geometric meaning of dot product

Front: The dot product satisfies $x^\top y = \|x\| \|y\| \cos \theta$. What does θ represent?

Back: The angle between vectors x and y

5. Condition for orthogonality

Front: Vectors x and y are orthogonal iff what condition holds?

Back: $x^\top y = 0$

6. Eigenvalue equation

Front: An eigenvector v of matrix A satisfies which defining equation?

Back: $Av = \lambda v$

7. Eigenvalues of symmetric matrices

Front: What key property do the eigenvalues of a real symmetric matrix satisfy?

Back: All eigenvalues are real

8. Unit eigenvectors

Front: How do you convert any eigenvector v into a **unit-norm** eigenvector?

Back: $v_{\text{unit}} = v / \|v\|$

9. Matrix rank

Front: What is the rank of a matrix?

Back: The number of linearly independent columns (or rows)

10. Frobenius norm

Front: The Frobenius norm of a matrix A is defined as $\|A\|_F = \sqrt{\sum_{i,j} \text{_____}}$. Fill in the blank.

Back: $|A_{ij}|^2$

If you want, the next batch can cover **probability prerequisites**, **calculus prerequisites**, **NumPy/tensor reasoning**, or another sub-area of linear algebra.



These are correct and accurately test and portray the basic probability knowledge needed for this class.

Now generate a second batch focusing only on probability prerequisites for CS 182: random variables, expectation, variance, conditional probability, and basic rules (product rule, Bayes' rule).

Produce exactly 10 high-quality flashcards that follow all the rules above

Here are 10 Anki-style flashcards focused only on probability prerequisites.

1. Random variable (discrete)

Front: A discrete random variable X is a function from the sample space to what set type?

Back: A **countable** set of possible values (e.g., integers)

2. Expectation (discrete)

Front: For a discrete random variable X with pmf $p(x)$, the expectation $E[X]$ is defined as

$E[X] = \sum_x \text{_____}$. Fill in the blank.

Back: $x p(x)$

3. Expectation (continuous)

Front: For a continuous random variable X with pdf $f(x)$, the expectation $E[X]$ is

$E[X] = \int_{-\infty}^{\infty} \text{_____} dx$. Fill in the blank.

Back: $x f(x)$

4. Variance definition

Front: The variance of a random variable X is defined as $\text{Var}(X) = E[\text{_____}]$. Fill in the blank.

Back: $(X - E[X])^2$

5. Variance shortcut formula

Front: Variance can be written in terms of moments as

$\text{Var}(X) = \text{_____} - (E[X])^2$. Fill in the blank.

Back: $E[X^2]$

6. Linearity of expectation

Front: For any random variables X, Y and constants a, b , what is $E[aX + bY]$?

Back: $aE[X] + bE[Y]$

7. Conditional probability definition

Front: For events A and B with $P(B) > 0$, the conditional probability $P(A | B)$ is defined as

$P(A | B) = \text{_____}$.

Back: $\frac{P(A \cap B)}{P(B)}$

8. Product rule

Front: Express the joint probability $P(A \cap B)$ using the product rule and conditional probability.

Back: $P(A \cap B) = P(A | B) P(B)$

9. Independence (events)

Front: Events A and B are independent iff what condition on their probabilities holds?

Back: $P(A \cap B) = P(A) P(B)$

10. Bayes' rule

Front: Bayes' rule relates $P(A | B)$ and $P(B | A)$. Write the formula for $P(A | B)$.

Back: $P(A | B) = \frac{P(B | A) P(A)}{P(B)}$ (for $P(B) > 0$)



Some of the cards are not very difficult, but this serves as a good starting point for the prompter to module what type of card and the difficulty level.

Now generate a third batch focusing only on calculus and gradients for CS 182: derivatives, gradients, chain rule, and the basic idea of gradient-based optimization.

Produce exactly 10 high-quality flashcards that follow all the rules above.

Here are 10 Anki-style flashcards focused only on calculus and gradients for CS 182.

1. Derivative as slope

Front: The derivative $f'(x)$ of a scalar function $f : \mathbb{R} \rightarrow \mathbb{R}$ at point x is the limit of what ratio?

Back: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

2. Gradient definition

Front: For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, what is the gradient $\nabla f(x)$?

Back: The vector of partial derivatives: $\nabla f(x) = \left[\frac{\partial f}{\partial x_1} \quad \dots \quad \frac{\partial f}{\partial x_n} \right]^T$

3. Direction of steepest ascent

Front: For a differentiable scalar function f , the gradient $\nabla f(x)$ points in which special direction?

Back: The direction of **steepest increase** of f at x

Good & well-scoped: This is a nice prereq-level chain rule card. It avoids heavy Jacobian notation and focuses on the scalar-intermediate case, which is appropriate for HW0 readiness and early CS 182 content.

4. Chain rule (single variable)

Front: If $y = f(u)$ and $u = g(x)$, what is $\frac{dy}{dx}$ in terms of f and g ?

Back: $\frac{dy}{dx} = \frac{df}{du} \cdot \frac{dg}{dx}$

5. Chain rule (vector form, scalar output)

Front: If $f : \mathbb{R}^m \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$, with $h(x) = f(g(x))$, then

$\nabla_x h(x) = \text{_____}^\top \nabla_u f(u) \big|_{u=g(x)}$. Fill in the blank.

Back: The Jacobian $J_g(x) : \nabla_x h(x) = J_g(x)^\top \nabla_u f(u) \big|_{u=g(x)}$

6. Gradient of quadratic form

Front: For $f(x) = x^\top A x$ with A symmetric, what is $\nabla_x f(x)$?

Back: $\nabla_x f(x) = 2Ax$

7. Gradient of squared norm

Front: For $f(x) = \frac{1}{2} \|x\|_2^2$, what is $\nabla_x f(x)$?

Back: $\nabla_x f(x) = x$

8. Gradient descent update rule

Front: In gradient descent on $f(\theta)$ with learning rate $\eta > 0$, what is the parameter update rule for θ ?

Back: $\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_\theta f(\theta_{\text{old}})$

9. Role of learning rate

Front: In gradient-based optimization, what is the effect of the learning rate η on updates?

Back: It scales the **step size** in the direction of the negative gradient

10. Stationary point condition

Front: For a differentiable scalar function $f(\theta)$, what condition characterizes a stationary point θ^* ?

Back: $\nabla_{\theta} f(\theta^*) = 0$



These cards are correct and very useful, and focus on concrete shaped and simple examples as requested in the prompt. The difficulty can be varied depending on the user's experience.

Now generate a fourth batch focusing only on Python / NumPy prerequisites for CS 182: arrays, shapes, broadcasting, basic tensor operations, and simple indexing.

Produce exactly 10 high-quality flashcards that follow all the rules above.

Here are 10 Anki-style flashcards focused only on Python / NumPy prerequisites: arrays, shapes, broadcasting, basic tensor operations, and simple indexing.

1. Creating a NumPy array

Front: In NumPy, which function is commonly used to create an array from a Python list?

Back: `np.array(...)`

2. Array shape

Front: In NumPy, the **shape** of an array `x` is accessed with which attribute?

Back: `x.shape`

3. Dot product vs. matmul (1D vectors)

Front: If `x` and `y` are 1D arrays of length `n`, what is the **shape** of `np.dot(x, y)` ?

Back: It is a **scalar** (0-dimensional)

4. Matrix–vector multiplication shape

Front: If `A` has shape `(m, n)` and `x` has shape `(n,)`, what is the shape of `A @ x` in NumPy?

Back: `(m,)`

5. Broadcasting rule (core idea)

Front: NumPy broadcasting aligns shapes starting from which end: leading dimensions or trailing dimensions?

Back: **Trailing** (rightmost) dimensions

6. Broadcasting example (scalar + vector)

Front: If `x` has shape `(5,)` and `b` is a scalar, what is the shape of `x + b` after broadcasting?

Back: `(5,)`

7. Axis meaning in reductions

Front: For a 2D array `X` of shape `(m, n)`, what is the shape of `X.sum(axis=0)` ?

Back: `(n,)`

8. Keepdims in reductions

Front: For a 2D array `X`, how do you sum along axis 1 **and keep** the reduced dimension as size 1?

Back: `X.sum(axis=1, keepdims=True)`

9. Basic 2D indexing

Front: For a 2D array `X`, what does `X[i, j]` return?

Back: The scalar element in **row** `i`, **column** `j`

10. Reshape constraint

Front: When calling `X.reshape(new_shape)`, what must be true about the total number of elements?

Back: It must stay the **same** before and after reshape



Borderline prereq: The Hessian and second-derivative test for convexity are useful for later in the course, but I wouldn't strictly classify them as "HW0 prerequisite knowledge." This feels more like early-CS-182 content than something students must already know on day one

Assume HW0 for CS 182 is mainly checking that students are ready for the course, not that they already know advanced deep learning.

From the entire set of cards you generated (including the rewrites), do the following:

1. Identify any cards that assume content that is clearly too advanced for HW0-level prereqs (for example, advanced deep learning topics that would not appear in a prereq check).
2. List those advanced or unrealistic cards and briefly explain why they are not appropriate as "Path to CS 182" prereq cards.
3. Propose corrected replacements for those cards that are more realistic HW0-style prereq content.

1. Cards that are too advanced for an HW0-style prereq check

From all four batches, essentially **one** card clearly goes beyond what you'd reasonably expect as a *prerequisite* for CS 182 HW0-level readiness:

- **Calculus batch – Card 5: “Chain rule (vector form, scalar output)”**

Front: If $f : \mathbb{R}^m \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$, with $h(x) = f(g(x))$, then $\nabla_x h(x) = \underline{\hspace{1cm}}^\top \nabla_u f(u) \big|_{u=g(x)}$. Fill in the blank.

Back: The Jacobian $J_g(x) : \nabla_x h(x) = J_g(x)^\top \nabla_u f(u) \big|_{u=g(x)}$

All other cards are basic linear algebra, probability, calculus, and NumPy/array skills that align well with a prereq check for HW0 (which itself reviews vector calculus and linear algebra, then builds up to ridge, SVD, ReLU, etc.).  hw0

2. Why this card is not appropriate as a “Path to CS 182” prereq card

- **Calculus Card 5 (Jacobian-based chain rule)**

- Uses **full Jacobian notation** $J_g(x)$ and a general mapping $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$.
- This is a *multi-variable, matrix-valued* chain rule that is usually *taught as part of* the course (or in more advanced vector calculus), rather than assumed as prereq.
- HW0 reviews vector derivatives and shapes but does **not** assume students already fluently manipulate Jacobians in general compositional settings; that's more like “early-CS-182 content” than “pre-CS-182 prereq”.  hw0

So it's slightly too “formal/general” for a readiness check, compared to something like “chain rule with a scalar intermediate variable” and basic gradient intuition.

3. Corrected replacement cards (more realistic HW0-style prereqs)

Here is a **replacement** card that keeps the spirit (chain rule with gradients) but stays at a simpler, HW0-prereq-friendly level:

Replacement for Calculus Card 5

Front:

If $u = g(x)$ with $g : \mathbb{R}^n \rightarrow \mathbb{R}$ and $y = f(u)$ with $f : \mathbb{R} \rightarrow \mathbb{R}$, and $h(x) = f(g(x))$, what is $\nabla_x h(x)$ in terms of f and g ?

Back:

$$\nabla_x h(x) = f'(g(x)) \nabla_x g(x)$$

No other cards need to be replaced for HW0-style “Path to CS 182” prerequisites.

Sources



A weakness of this is the possibility that, due to ChatGPT's limitations, it won't accurately be able to churn out a plethora of diverse and high-quality cards, so constant iteration and segmenting is required, along with user overview.

Now check coverage against this target list of prereq topics:

- Linear algebra: vectors, matrices, norms, dot products, eigenvalues/eigenvectors
- Probability: random variables, expectation, variance, conditional probability, basic rules (product rule, Bayes' rule)
- Calculus: derivatives, gradients, chain rule, basic gradient-based optimization idea
- Python/NumPy: arrays, shapes, broadcasting, basic tensor ops, simple indexing

1. For each of the four groups above, tell me briefly whether my current deck is missing or underrepresenting important concepts.
2. For any group that is missing important ideas, propose 3–5 new cards that fill those gaps.

Below is a concise coverage check, followed by gap-filling cards only for the topics that need strengthening.

1. Coverage Assessment by Topic Group

Linear Algebra (vectors, matrices, norms, dot products, eigenvalues/eigenvectors)

Coverage is **good** but **underrepresents**:

- Matrix norms beyond Frobenius (optional but helpful)
- Basic matrix operations (transpose, identity, invertibility)
- Orthogonal/orthonormal matrices
- Properties of eigenvectors/eigenvalues relevant to ML (e.g., eigenbasis of symmetric matrices)

Probability (random variables, expectation, variance, conditional probability, rules)

Coverage is **strong**; only **minor gaps**:

- Law of total probability
- Conditional expectation definition
- Covariance (basic prerequisite for multivariate thinking)

Calculus (derivatives, gradients, chain rule, gradient-based optimization)

Coverage is **good** but **missing**:

- Hessian (basic idea, not computation)
- Conditions for convexity (very simple scalar case)
- Multivariate chain rule intuition (scalar intermediate functions already fixed)
- Gradient descent convergence intuition (high-level, prereq-friendly)

Python / NumPy (arrays, shapes, broadcasting, tensor ops, indexing)

Coverage is **good** but **missing**:

- Boolean masking
- Slicing semantics (e.g., `:` behavior)
- `np.newaxis` / reshape basics
- Elementwise vs. matrix operations clarity
- Common reduction ops beyond sum (mean, max)

2. Gap-Filling Cards (3–5 per underrepresented group)

Linear Algebra — Additional Prereq Cards

LA-1. Identity matrix definition

Front: What is the defining property of the identity matrix I_n when multiplied by a vector $x \in \mathbb{R}^n$?

Back: $I_n x = x$

LA-2. Condition for invertibility

Front: A square matrix A is invertible iff what condition on its determinant holds?

Back: $\det(A) \neq 0$

LA-3. Orthonormal columns

Front: A matrix Q has orthonormal columns iff $Q^T Q = \underline{\hspace{2cm}}$. Fill in the blank.

Back: I

LA-4. Eigenbasis for symmetric matrices

Front: Real symmetric matrices admit what special type of eigenvector basis?

Back: An **orthonormal** eigenvector basis

LA-5. Transpose property for dot product

Front: For vectors x, y , how can the dot product be written using a transpose?

Back: $x^T y$

Probability — Additional Prereq Cards

PR-1. Law of total probability

Front: For a partition $\{B_i\}$ with $P(B_i) > 0$,

$P(A) = \sum_i \underline{\hspace{2cm}}$. Fill in the blank.

Back: $P(A \mid B_i) P(B_i)$

PR-2. Definition of covariance

Front: The covariance of random variables X, Y is

$\text{Cov}(X, Y) = E[\underline{\hspace{2cm}}]$. Fill in the blank.

Back: $(X - E[X])(Y - E[Y])$

PR-3. Conditional expectation (discrete)

Front: For discrete X , $E[X \mid Y = y]$ is defined as

$\sum_x$ _____. Fill in the blank.

Back: $x P(X = x \mid Y = y)$

PR-4. Variance of scaled variable

Front: For constant a , $\text{Var}(aX) =$ _____.

Back: $a^2 \text{Var}(X)$

Calculus & Gradients — Additional Prereq Cards

CA-1. Hessian definition (basic idea)

Front: The Hessian $H_f(x)$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a matrix whose (i, j) entry is $\frac{\partial^2 f}{\partial x_i \partial x_j}$.

What does this matrix represent?

Back: Second-order curvature of f

CA-2. Convexity (scalar second-derivative test)

Front: A twice-differentiable scalar function $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex iff what condition on $f''(x)$ holds?

Back: $f''(x) \geq 0$ for all x

CA-3. Gradient of affine function

Front: If $f(x) = a^\top x + b$, what is $\nabla_x f(x)$?

Back: a

CA-4. Gradient descent intuition

Front: Gradient descent updates in the direction of $-\nabla f$ because this direction does what to the function value locally?

Back: Decreases it most rapidly

CA-5. Chain rule with scalar intermediate

Front: If $u = g(x)$ and $y = f(u)$, what is $\nabla_x y$?

Back: $f'(g(x)) \nabla_x g(x)$

Python / NumPy — Additional Prereq Cards

NP-1. Slicing with ':'

Front: In NumPy, what does `x[:, j]` select from a 2D array?

Back: All rows, column `j`

NP-2. Boolean masking

Front: If `mask` is a boolean array of the same shape as `x`, what does `x[mask]` return?

Back: The elements of `x` where the mask is `True`

NP-3. Using `np.newaxis`

Front: What does inserting `np.newaxis` into an index do to the array's shape?

Back: Adds a new dimension of size 1

NP-4. Elementwise vs. matrix multiply

Front: In NumPy, which operator performs elementwise multiplication: `*` or `@`?

Back: `*`

NP-5. Common reduction operations

Front: Name two common NumPy reductions besides `sum`.

Back: `mean`, `max` (also `min`, `std`)

If desired, I can now generate a **fifth batch of cards** incorporating only the missing concepts identified above, or revise your full deck into a polished prereq Anki package.



Now produce a final cleaned-up deck suitable for importing into Anki.

Constraints:

- At most 40 total cards.
- All cards must follow the flashcard rules (active recall, one idea per card, concise, cloze deletions when useful).
- The deck should have balanced coverage across:
 - Linear algebra
 - Probability
 - Calculus/gradients
 - Python/NumPy

Below is a **40-card cleaned, balanced deck** ready for Anki import (Q/A plain text).
All cards follow active recall, minimum-information, and cloze rules.

FINAL 40-CARD PREREQ DECK FOR CS 182

LINEAR ALGEBRA (10 cards)

LA1 — ℓ_2 norm

Q: What is the definition of the ℓ_2 norm of a vector $x \in \mathbb{R}^n$?

A: $\|x\|_2 = \sqrt{\sum_i x_i^2}$

LA2 — Dot product squared-norm

Q: The expression $x^\top x$ equals what?

A: $\|x\|_2^2$

LA3 — Orthogonality condition

Q: Vectors x and y are orthogonal iff _____.

A: $x^\top y = 0$

LA4 — Matrix–vector product shape

Q: If $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$, what is the shape of Ax ?

A: $m \times 1$

LA5 — Identity matrix property

Q: What is the defining property of the identity matrix I_n when applied to a vector x ?

The final flashcards have some clarity issues when it comes to under-specifying solutions. Most are done well, but for LA2, I'd write the front as "For a vector x , what is $x^\top x$ equal to?" so I know exactly which expression is being referred to.]

A: $I_n x = x$

LA6 — Orthonormal columns

Q: A matrix Q has orthonormal columns iff $Q^\top Q = \underline{\hspace{2cm}}$.

A: I

LA7 — Eigenvalue equation

Q: An eigenvector v of A satisfies which equation?

A: $Av = \lambda v$

LA8 — Symmetric matrix eigenvalues

Q: What special property do the eigenvalues of a real symmetric matrix have?

A: They are all real.

LA9 — Eigenbasis of symmetric matrices

Q: Real symmetric matrices admit what special kind of eigenvector basis?

A: An orthonormal eigenbasis.

LA10 — Condition for invertibility

Q: A square matrix A is invertible iff $\det(A) \underline{\hspace{2cm}}$.

A: $\det(A) \neq 0$

PROBABILITY (10 cards)

PR1 — Expectation (discrete)

Q: For discrete X with pmf $p(x)$,

$$\mathbb{E}[X] = \sum_x \underline{\hspace{2cm}}.$$

A: $x p(x)$

PR2 — Expectation (continuous)

Q: For continuous X with pdf $f(x)$,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \underline{\hspace{2cm}} dx.$$

A: $x f(x)$

PR3 — Variance definition

Q: $\text{Var}(X) = \mathbb{E}[(\underline{\hspace{2cm}})^2]$.

A: $X - \mathbb{E}[X]$

PR4 — Variance shortcut

Q: $\text{Var}(X) = \mathbb{E}[X^2] - \underline{\hspace{2cm}}$.

A: $(\mathbb{E}[X])^2$

PR5 — Conditional probability

Q: $P(A \mid B) = \frac{P(A \cap B)}{\underline{\hspace{2cm}}}$.

A: $P(B)$

PR6 — Product rule

Q: The product rule states $P(A \cap B) = P(A \mid B) \underline{\hspace{2cm}}$.

A: $P(B)$

PR7 — Bayes' rule

Q: $P(A \mid B) = \frac{P(B \mid A)P(A)}{\underline{\hspace{2cm}}}$.

A: $P(B)$

PR8 — Law of total probability

Q: For a partition $\{B_i\}$,

$P(A) = \sum_i \underline{\hspace{2cm}}$.

A: $P(A \mid B_i)P(B_i)$

PR9 — Covariance definition

Q: $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(\underline{\hspace{2cm}})]$.

A: $Y - \mathbb{E}[Y]$

PR10 — Conditional expectation (discrete)

Q: $\mathbb{E}[X \mid Y = y] = \sum_x \underline{\hspace{2cm}}$.

A: $x P(X = x \mid Y = y)$

CALCULUS / GRADIENTS (10 cards)**CA1 — Derivative definition**

Q: $f'(x) = \lim_{h \rightarrow 0} \frac{\underline{\hspace{2cm}}}{h}$.

A: $f(x + h) - f(x)$

CA2 — Gradient definition

Q: For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, what is $\nabla f(x)$?

A: The vector of partial derivatives.

CA3 — Gradient of squared norm

Q: If $f(x) = \frac{1}{2} \|x\|_2^2$, then $\nabla_x f(x) = \underline{\hspace{2cm}}$.

A: x

CA4 — Gradient of affine function

Q: If $f(x) = a^\top x + b$, what is $\nabla_x f(x)$?

A: a

CA5 — Chain rule (scalar intermediate)

Q: If $u = g(x)$, $y = f(u)$, what is $\nabla_x y$?

A: $f'(g(x)) \nabla_x g(x)$

CA6 — Direction of steepest ascent

Q: The gradient $\nabla f(x)$ points in what direction relative to f ?

A: Steepest increase.

CA7 — Gradient descent update

Q: Gradient descent uses the update $\theta \leftarrow \theta - \eta \underline{\hspace{2cm}}$.

A: $\nabla_\theta f(\theta)$

CA8 — Role of learning rate

Q: In gradient descent, the learning rate η controls what aspect of updates?

A: Step size.

CA9 — Hessian meaning

Q: The Hessian $H_f(x)$ represents second-order information about what property of f ?

A: Its curvature.

CA10 — Convexity (scalar)

Q: A twice-differentiable scalar function is convex iff $f''(x) \underline{\hspace{2cm}}$.

A: $f''(x) \geq 0$

PYTHON / NUMPY (10 cards)

NP1 — Creating arrays

Q: Which function converts a Python list into a NumPy array?

A: `np.array(...)`

NP2 — Array shape

Q: The shape of a NumPy array `x` is accessed with `x.shape`.

A: `shape`

NP3 — Matrix–vector multiply shape

Q: If `A` has shape (m, n) and `x` has shape $(n,)$, then `A @ x` has shape ____.

A: $(m,)$

NP4 — Broadcasting alignment

Q: NumPy broadcasting aligns array shapes starting from which side?

A: Trailing (rightmost) dimensions.

NP5 — Slicing with ":"

Q: `x[:, j]` selects what part of a 2D array?

A: All rows, column `j`.

NP6 — Boolean masking

Q: If `mask` is boolean and same shape as `x`, what does `x[mask]` return?

A: Elements where `mask` is `True`.

NP7 — `np.newaxis`

Q: Inserting `np.newaxis` into an index adds what to the shape?

A: A new dimension of size 1.

NP8 — Elementwise vs matrix multiply

Q: Which operator performs elementwise multiplication in NumPy?

A: `*`

NP9 — Common reductions

Q: Name a NumPy reduction besides `sum`.

A: `mean` or `max` (either acceptable)

NP10 — Reshape constraint

Q: `x.reshape(new_shape)` requires what to stay unchanged?

A: Total number of elements.
